Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 06-05-2026**

Time: 09.00Am to 12.00Pm **GTA-30** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**B**	2	**C**	3	**B**	4	**C**	5	**AC**	6	**AD**
7	**CD**	8	**381**	9	**64**	10	**4**	11	**1**	12	**8**
13	**2**	14	**C**	15	**D**	16	**C**	17	**A**		

PHYSICS

18	**A**	19	**C**	20	**B**	21	**C**	22	**AC**	23	**ABCD**
24	**BD**	25	**450**	26	**37**	27	**4**	28	**125**	29	**15**
30	**10**	31	**A**	32	**A**	33	**D**	34	**B**		

CHEMISTRY

35	**D**	36	**C**	37	**B**	38	**C**	39	**ABC**	40	**ABD**
41	**ABD**	42	**4**	43	**40**	44	**2**	45	**920**	46	**1432**
47	**187**	48	**D**	49	**D**	50	**B**	51	**D**		

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SOLUTIONS
MATHEMATICS

1. The square in the first step has area $4\cdot 4=16$. There are four squares added in the second step, each with area $1\cdot 1=1$ for a total of 4 . After this, there are 3 times as many squares added in each step, and each square has area $\frac{1}{4}\cdot\frac{1}{4}=\frac{1}{16}$ as large as the areas of the squares added in the previous step. Thus, each step adds $\frac{3}{16}$ the area that was added in the previous step. The total area of all the squares added in every step is

$$16+4+4\cdot\frac{3}{16}+4\left(\frac{3}{16}\right)^2+4\left(\frac{3}{16}\right)^3+\cdots.$$

This is 16 plus an infinite geometric series with first term 4 and common ratio $\frac{3}{16}$ which sums to $16+4\cdot\frac{1}{1-\frac{3}{16}}=16+\frac{64}{13}=\frac{272}{13}$.

The requested sum is $272+13=285$

2. Let $0<x<y<z\le 8; x$ cannot be 1 because it makes $xy>z\to y>z$; $x=2, y=3, z$ can be 4,5 but not others; $x=2, y=4, z$ can be $5,6,7; x=2, y=5, z$ can be 6, 7, $8; x=2, y=6, z$ can be 7, $8; x=2, y=7, z$ can be 8 ; for $x=2$, total 11 cases; Similarly, for $x=3, y=4,5,6,7$, total 10 cases; for $x=4, y=5,6,7$, total 6 cases; $x=5, y=6$, 7, 3 cases; $x=6, y=7, z=8,1$ cases; Total $=11+10+6+3+1=31$.

Per mutate x, y, z for ordered triple, it is $31\cdot 6=186$

3. Let $h(x)=f^{-1}(g^{-1}(x))$

$$h'(x)=(f^{-1})(g^{-1}(x))(g^{-1}(x))^1$$

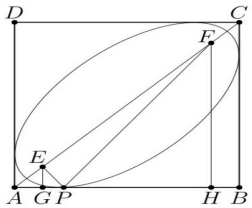
$$h'(3)=(f^{-1})^1(g^{-1}(3))(g^{-1}(3))^1$$

$$h'(3)=(f^{-1}(1))^1(g^{-1}(3))^{-1}$$

$$=\frac{1}{f'(f^{-1}(1)}\times\frac{1}{g'(g^{-1}(3))}$$

$$=\frac{1}{f'(2)}\times\frac{1}{g'(1)} \qquad =\frac{1}{4}\times\frac{1}{5}=\frac{1}{20}$$

4.



$2a\ell=18\sqrt{2}, b^2=63 \qquad \therefore\ a=15.$

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5. Let $f(x)$ be any continuous even function on $[-1,1]$. We have the particular case of $f(x)=x^4+|x|$ but we will not use that formula in detail - only the fact that it is even.

Let $I_1=\int_0^{\pi} f(\cos x)\,dx$ and $I_2=\int_0^{\frac{\pi}{2}} f(\sin x)\,dx$.

Now make the substitution $u=\cos x$ in the first integral and $u=\sin x$ in the second.

$I_1=\int_{-1}^{1}\frac{f(u)}{\sqrt{1-u^2}}du$ and $I_2=\int_0^1\frac{f(u)}{\sqrt{1-u^2}}du$

Since $\frac{f(u)}{\sqrt{1-u^2}}$ is an even function of u, it is clear by symmetry that $I_1=2I_2$.

6. $AP=\sqrt{(1-\cos\alpha)^2+\sin^2\alpha}=2\left|\sin\frac{\alpha}{2}\right|=2\sin\frac{\alpha}{2}$ $\left(\because \sin\frac{\alpha}{2}>0\right)$

Similarly, $AQ=2\sin\frac{\beta}{2}$ and $AR=2\sin\frac{\gamma}{2}$

Now AP, AQ, AR are in G.P. $\Rightarrow \sin\frac{\alpha}{2},\sin\frac{\beta}{2},\sin\frac{\gamma}{2}$ are in G.P.

$$\therefore \frac{\sin\frac{\alpha}{2}+\sin\frac{\gamma}{2}}{2}\geq\sin\frac{\beta}{2}\Rightarrow\sin\frac{\alpha+\gamma}{4}\cos\frac{\alpha-\gamma}{4}\geq\sin\frac{\beta}{2}$$

Also, $\sin\frac{\alpha}{2}\sin\frac{\gamma}{2}\leq\sin\frac{\beta}{2}$.

Now $\frac{1\left(x+y\frac{dy}{dx}\right)+2(0)}{1+2}=x\Rightarrow x+y\frac{dy}{dx}=3x\ \therefore y\frac{dy}{dx}=2x\#(1)$

$\Rightarrow\int ydy=\int 2xdx\Rightarrow\frac{y^2}{2}=x^2+C$

Also $(0,4)$ satisfy it, so $C=8$

$\therefore y^2=2x^2+16$ (equation of curve) which represent a hyperbola.

Also $\left.\frac{dy}{dx}\right]_{(4+\sqrt{5})}=\frac{2(4)}{4\sqrt{3}}=\frac{2}{\sqrt{3}}$

$\therefore$ The equation of tangent at $\left(4,4\sqrt{3}\right)$ is $y-4\sqrt{3}=\frac{2}{\sqrt{3}}(x-4)\Rightarrow 2x-\sqrt{3}y+4=0$

7. $\left(x^2+(y+1)^2\right)ydx+\left(x^2+x\right)dy=(x+1)(y+1)\left((x+1)dx+(y+1)dy\right)$

$\Rightarrow\left(x^2+(y+1)^2\right)ydx+\left(x^2+x+x^3+x^2-x^3-x^2\right)dy$

$=(x+1)^2(y+1)dx+(x+1)(y+1)^2dy\Rightarrow\left(x^2+(y+1)^2\right)ydx+x(x+1)^2dy$

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$$=(x+1)^2(y+1)dx+(x+1)\left((y+1)^2+x^2\right)dy$$

$$\left(x^2+(y+1)^2\right)\left(ydx-(x+1)dy\right)=(x+1)^2((y+1)-xdy)$$

$$\frac{-\left((x+1)dy-ydx\right)}{(x+1)^2}=\frac{(y+1)dx-xdy}{x^2+(y+1)^2}-d\left(\frac{y}{x+1}\right)=d\left(\tan^{-1}\left(\frac{x}{y+1}\right)\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{x}{y+1}\right)+\left(\frac{y}{x+1}\right)=c$$

Passes through origin $\Rightarrow C=0\Rightarrow \tan^{-1}\left(\frac{x}{y+1}\right)+\left(\frac{y}{x+1}\right)=0$

Passes through (a,b) and $a=b+1\Rightarrow a=\frac{4-\pi}{4+\pi},b=\frac{-2\pi}{4+\pi}$

8. Note that the arithmetic becomes much simpler if each element of the set is reduced by 107 and then divided by 5 to give the set $\{0,3,4,6,9,y\}$, where $y=\frac{x-107}{5}$. The mean of this set is $\frac{22+y}{6}$. There are three cases.

- If $y<3$, then the median of the set is $\frac{3+4}{2}=\frac{7}{2}$. Thus, $\frac{22+y}{6}=\frac{7}{2}$, so $y=-1$.
- If $3\le y\le 6$, then the median of the set is $\frac{4+y}{2}$. Thus, $\frac{22+y}{6}=\frac{4+y}{2}$, so $y=5$.
- If $6<y$, then the median of the set is $\frac{4+6}{2}=5$. Thus, $\frac{22+y}{6}=5$, so $y=8$.

Therefore, the three possible values for x are $5\cdot(-1)+107=102, 5\cdot5+107=132$, and $5\cdot8+107=147$, and the sum of the possible values of x is $102+132+147=381$

9. We have $f(\alpha)=\int_{\alpha^{-1}}^{\alpha}\frac{1}{x}\cot^{-1}\left(\frac{x^2-x+1}{2x-3x^2}+\frac{x^2-x+1}{3-2x}\right)dx\quad x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$

$$f(\alpha)=\int_{\frac{1}{\alpha}}^{\alpha}t\cot^{-1}\left(\frac{t^2-t+1}{2t-3}+\frac{t^2-t+1}{2t-3t^2}\right)\left(\frac{-1}{t^2}\right)dt$$

$$=\int_{\frac{1}{\alpha}}^{\alpha}\frac{1}{t}\cot^{-1}\left(\frac{t^2-t+1}{2t-3}+\frac{t^2-t+1}{3t^2-2t}\right)dt \quad =\int_{\frac{1}{\alpha}}^{\alpha}\frac{1}{t}\left\{\pi-\cot^{-1}\left(\frac{t^2-t+1}{3-2t}+\frac{t^2-t+1}{2t-3t^2}\right)\right\}dt$$

$\therefore$ On equation (1) + equation (2), we get

$$2f(\alpha)=\int_{\frac{1}{\alpha}}^{\alpha}\frac{\pi}{t}=\pi\left(\ell n\alpha-\ell n\left(\frac{1}{\alpha}\right)\right)=2\pi\ell n\alpha\quad g(\alpha)=\int_{\ln\left(\frac{1}{\alpha}\right)}^{\ln\alpha}1.dx=\ell n\alpha-\ell n\left(\frac{1}{\alpha}\right)=2\ell n\alpha$$

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Now $f(200)-\frac{\pi}{2}g(50)=\pi\ell n(200)-\pi\ell n(50)=\pi\ell n4=3\cdot\frac{\pi}{3}\ell n4$

Hence $b^a=4^3=64$

10. Let $P(4+4\cos\theta,3+3\sin\theta)$ be a paint on given elipse.

Its image in line $x-y-2=0$ is $(5+3\sin\theta,2+4\cos\theta)$.

11. $\int_4^8\frac{f'(x)dx}{f^2(x)}=2$

Now, $\int_4^8\left(\frac{f'(x)}{(f(x))^2}+k\right)^2dx=\int_4^8\frac{(f'(x))^2}{(f(x))^4}dx+2k\int_4^8\frac{f'(x)}{(f(x))^2}dx+k^2\int_4^8dx=(2k+1)^2$

Now, i. e. $\int_4^8\left(\frac{f'(x)}{(f(x))^2}-\frac{1}{2}\right)^2dx=0$

$\Rightarrow\frac{f'(x)}{(f(x))^2}=\frac{1}{2}\Rightarrow-\frac{1}{f(x)}=\frac{x}{2}+c-4=2+c;c=-6\frac{1}{f(x)}=-\frac{x}{2}+6;\frac{1}{f(6)}=-3+6=3f(6)=\frac{1}{3}$

12. The numerator can be written as $2^{2a}+2^{2a+2k}+\ldots+2^{2a+2mk}$, which is actually a sum of geometric series. This can be expressed as $2^{2a}\cdot\frac{1-2^{2k(m+1)}}{1-2^{2k}}$. The denominator in the same way can be expressed as $2^a\cdot\frac{1-2^{k(m+1)}}{1-2^k}$.

Doing some algebra on the top and bottom we get: $2^a\cdot\frac{1+2^{k(m+1)}}{1+2^k}=964$

The prime factorization of 964 is $241\cdot2^2$. Equating $2^a=2^2$, we get $a=2$.
Next since 241 is a prime number we equate the latter half of the product to 241.

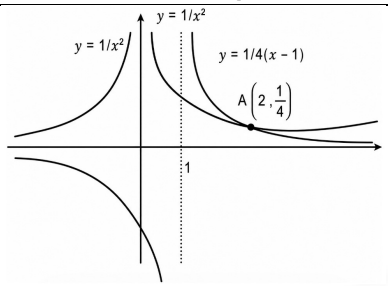
$\frac{1+2^{k(m+1)}}{1+2^k}=241$

Doing some algebra we get, that $2^k\left(2^{km}-241\right)=240$

The closest power of 2 to 241 is 256 which is 2^8. So setting $km=8$,
we get $2^8-241=15=240,2^k=16,k=4$.
This we know that if $k=4$, then $4m=8$, so $m=2$.
Finally, we conclude that $a=2,m=2$, and $k=4.4+2+2=8$, so the answer is A

13.

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$y_1 = \frac{1}{x^2}$ and $y_2 = \frac{1}{4(x-1)}$

These curves are touching at $x = 2$.

$A = \int_2^a \left(-\frac{1}{x^2} + \frac{1}{4(x-1)} \right) dx = \frac{1}{x} + \frac{1}{4}\ln(x-1)\Big|_2^a$ or $\left(\frac{1}{a} + \frac{1}{4}\ln(a-1) - \frac{1}{2}\right) = \frac{1}{a}$

or $\ln(a-1) - 2 = 0 \Rightarrow a = 1 + e^2$

Also, $A = \int_b^2 \left(-\frac{1}{x^2} + \frac{1}{4(x-1)} \right) dx = 1 - \frac{1}{b}$

or $\frac{1}{2} - \frac{1}{4}\ln(b-1) - \frac{1}{b} = 1 - \frac{1}{b}$ or $b = 1 + 1/e^2$

14. $\frac{n}{1-3n^2+n^4} = \frac{n}{\left(n^2-1\right)^2 - n^2} = \frac{\left(n^2-1+n\right) - \left(n^2-1-n\right)}{2\left(n^2-1-n\right)\left(n^2-1+n\right)}$

$\frac{1}{R_n} - \frac{1}{R_{n+1}} = \frac{1}{2\left(n^2-n-1\right)} - \frac{1}{2\left(n^2+n-1\right)}$

Summing the above from 1 to $n-1$ and using the value $R_1 = -2$, we get:

$R_n = 2\left(n^2 - n - 1\right)$

(a) $R_n > 0$

$\Rightarrow n \in \left(-\infty, \frac{1-\sqrt{5}}{2}\right) \cup \left(\frac{1+\sqrt{5}}{2}, \infty\right) \Rightarrow n = \{2,3,4,5,\ldots\}$

$\Rightarrow (p),(q),(r),(s)$

(b) $S_n = 2\left[2\left(n^2+4n+4-n-2-1\right) - 3\left(n^2-n-1\right)\right] = 2\left[-n^2+9n+5\right]$

S_n is maximum at $n = 4,5$ $\Rightarrow (r),(s)$

15. $S = \{(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(4,2),(5,2),(6,2)\}$

(I) $\rightarrow m = 3$ $R = \{(3,1),(3,2),(3,3),(1,3),(2,3)\}$ $P\left(\frac{R}{S}\right) = \frac{2}{9}$

(II) $\rightarrow m = 4$ $R = \{(4,1),(4,2),(4,3),(4,4),(1,4),(2,4),(3,4)\}$ $P\left(\frac{R}{S}\right) = \frac{2}{9}$

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(iii) $\rightarrow m=5$ $R=\{(5,1),(5,2),(5,3),(5,4),(5,5),(1,5),(2,5),(3,5),(4,5)\}$ $P\left(\frac{R}{S}\right)=\frac{2}{9}$

(iv) $\rightarrow m=6$ $R=\{(6,1),(6,2),(6,3),(6,4),(6,5),(6,6),(1,6),(2,6),(3,6),(4,6),(5,6)\}$

$P\left(\frac{R}{S}\right)=\frac{2}{9}$

16. Given, $L:\frac{x-1}{1}=\frac{y-1}{-1}=\frac{z-1}{-1}=\lambda$ (say)

$\therefore$ Any point on it is $(1+\lambda,1-\lambda,1-\lambda)$

As, P_1 is the plane containing L and perpendicular to $\vec{v}_1=\hat{i}+\hat{j}$, so equation of P_1 is

$1(x-1)+1(y-1)+c(z-1)=0\Rightarrow x+y=2\#(1)$

Let equation of plane P_2 is

$a(x-1)+b(y-1)+c(z-1)=0\#(3)$

Where $a-b-c=0\#(2)$

Also, $(0,1,0)$ lies on plane p_2, so $-a-c=0$

$\therefore$ From (2) and (3),

$\frac{a}{1}=\frac{b}{2}=\frac{c}{-1}\Rightarrow$ equation of plane P_2 is

$1(x-1)+2(y-1)-1(z-1)=0\Rightarrow x+2y-z=2\#(4)$

Distance of plane $x+y-2=0$ from $\mathrm{M}(1,2,3)$ is $=\frac{|1+2-2|}{\sqrt{2}}=\frac{1}{\sqrt{2}}$. Ans.

Let $\mathrm{B}(1+\lambda,1-\lambda,1-\lambda)$. Also, A is $(-1,2,-1)$

$\therefore \overrightarrow{\mathrm{AB}}=$ p.v. of B $-$ p.v. of A $=(2+\lambda,-1-\lambda,2-\lambda)$

As $\overrightarrow{\mathrm{AB}}$ is parallel to plane $2x+y+3z=1$, so

$2(2+\lambda)+(-1-\lambda)+3(2-\lambda)=0\Rightarrow(4+2\lambda)+(-1-\lambda)+(6-3\lambda)=0$

$\Rightarrow 2\lambda=9\Rightarrow\lambda=\frac{9}{2}$. Ans.

Angle between P_1 and P_2, is

$$\cos\theta=\frac{\vec{n}_1\cdot\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}=\frac{1(1)+1(2)+(0)(-1)}{\sqrt{2}\sqrt{6}}=\frac{3}{\sqrt{2}\sqrt{6}}=\frac{3}{\sqrt{2}\sqrt{3}\sqrt{2}}=\frac{\sqrt{3}}{2}$$

17. Since $\frac{x^n}{1+x^2}+\frac{x^{n-2}}{1+x^2}=x^{n-2}$, one has $a_n+a_{n-2}=\frac{1}{n-1}$ for $n\geq 2$. Also $a_1=\frac{1}{2}\ln 2$. Hence

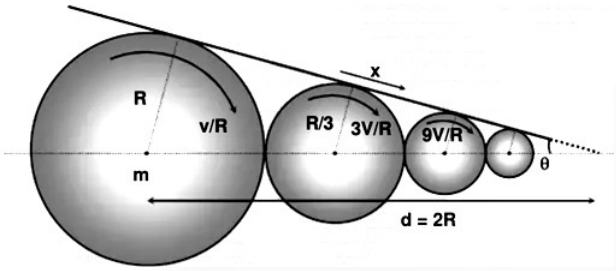
$$a_3=\frac{1}{2}-\frac{\ln 2}{2}=\frac{1-\ln 2}{2},\ a_1+a_3=\frac{1}{2},\ \text{and}$$

$$a_5+2a_3+a_1=(a_5+a_3)+(a_3+a_1)=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}.$$

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PHYSICS

18.



$$d = R + \frac{2R}{3} + \frac{2R}{9} + \ldots\ldots = R + \frac{2R}{3}\left(1 + \frac{1}{3} + \ldots\ldots\right) = R + \frac{2R}{3}\left(\frac{1}{1-\frac{1}{3}}\right) = 2R$$

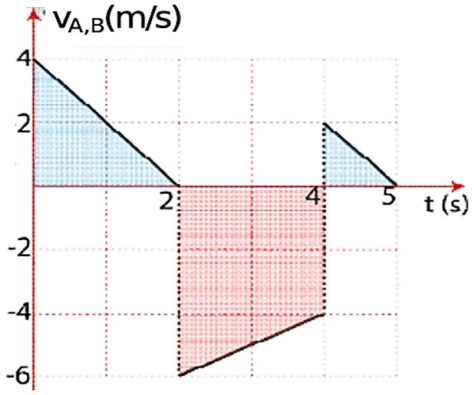
$$\sin\theta = \frac{R}{d} = \frac{R}{2R} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

$$mgx\sin 30^\circ = \frac{1}{2}mv^2 + \frac{1}{2}\left[\frac{mR^2}{2}\left(\frac{v}{R}\right)^2 + \frac{1}{2}\left(\frac{m}{9}\right)\left(\frac{R}{3}\right)^2\left(\frac{3v}{R}\right)^2 + \frac{1}{2}\left(\frac{m}{81}\right)\left(\frac{R}{9}\right)^2\left(\frac{9v}{R}\right)^2\right]$$

$$mgx\left(\frac{1}{2}\right) = \frac{mv^2}{2} + \frac{1}{4}mv^2\left[1 + \frac{1}{3^2} + \frac{1}{3^4} + \frac{1}{3^6} + \ldots\ldots\right] \quad gx = \frac{25v^2}{16}$$

Differentiate $g\left(\frac{dx}{dt}\right) = \frac{25v}{8}\left(\frac{dv}{dt}\right) \Rightarrow a = \frac{8g}{25}$

19. Following graph shows the relative velocity of 'A' w.r.t ' B '. Area under the graph represents the separation between the particles.



At $t = 2$ s, separation between the particles $= \frac{1}{2} \times 2 \times 4 = 4$ m

At $t = 4$ s, separation between the particles $= \left|4 - \frac{1}{2}(6+4) \times 2\right| = 6$ m

$=$ At $t = 5$ s, separation between the particles $= \left|-6 \text{ m} + \frac{1}{2} \times 1 \times 2\right| = 5$ m

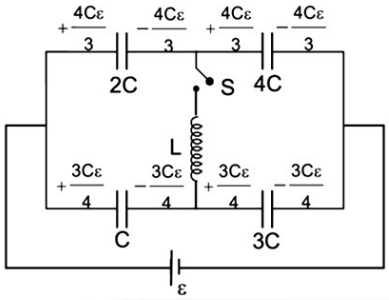
Hence, the maximum separation between the particles is 6 m.

20. Before the switch 'S' is closed.

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$$C_{eq} = \frac{4C}{3} + \frac{3C}{4} = \frac{25C}{12}$$

$$Q = C_{eq}\varepsilon = \frac{25C\varepsilon}{12}$$

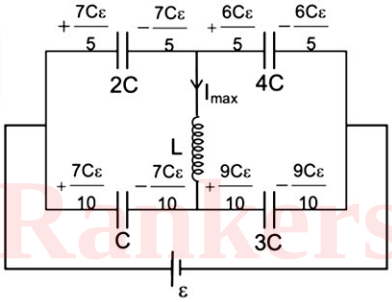


When the current through the inductor is maximum after the switch 'S' is closed.

$$C'_{eq} = \frac{3C \times 7C}{10C} = \frac{21C}{10}$$

$$Q' = C'_{eq} = \frac{21C\varepsilon}{10}$$

Charge flown through the battery after the switch 'S' is closed.



$$\Delta Q = \left(\frac{21C\varepsilon}{10}\right) - \left(\frac{25C\varepsilon}{12}\right)$$

$$\Delta Q = \frac{C\varepsilon}{60}$$

Now, $\Delta W_b = \Delta U + \frac{1}{2}LI_{max}^2$

$$\frac{C\varepsilon^2}{60} = \frac{1}{2}\left(\frac{21C}{10} - \frac{25C}{12}\right)\varepsilon^2 + \frac{1}{2}LI_{max}^2$$

$$\Rightarrow \frac{C\varepsilon^2}{120} = \frac{1}{2}LI_{max}^2$$

$$\frac{C\varepsilon^2}{60} = \frac{C\varepsilon^2}{120} + \frac{1}{2}LI_{max}^2$$

$$\Rightarrow I_{max} = \frac{\varepsilon}{2}\sqrt{\frac{C}{15\,L}}$$

21. Let mesh currents be:

- corner squares: a
- side-middle squares: b
- center square: c

Branch XY lies between:

- top-middle square current b
- center square current c

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Hence current through XY is

$$I_{XY} = c - b$$

Now look at the central square.

All 4 surrounding side-middle squares are identical, so on each side of the center square the opposing neighbour current is b.

Thus for the central square:

$4r(c-b) = 4\varepsilon_0$

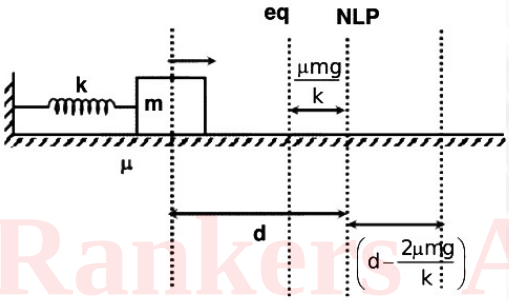
So immediately,

$$c - b = \frac{\varepsilon_0}{r}$$

Therefore

$$\boxed{I_{XY} = \frac{\varepsilon_0}{r}}$$

22. If we compress the spring by d from NLP then it comes to rest first time at a distance $\left(d - \frac{2\mu mg}{k}\right)$ from NLP on the other side.



Finally we want block to stop at NLP

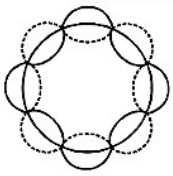
$$d - \left(\frac{2\mu mg}{k}\right)n = 0$$

$$d = \left(\frac{2\mu mg}{k}\right)n$$

Here, $n = 1,2,3,4,\ldots\ldots$

(NLP = natural length position of spring)

23. $\frac{8\lambda}{2} = 2\pi R$



$4\lambda = 2\pi R \Rightarrow n\lambda$

$n = 4$

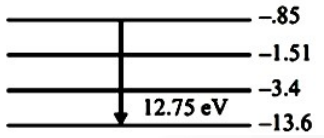
So given excite state $= n = 4$.

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(A) Change in angular momentum $= \frac{(n_2 - n_1)h}{2\pi} = \frac{(4-1)\pi}{2\pi} = \frac{3h}{2\pi}$

(B) $n\lambda = 2\pi R_n \qquad \lambda = 2\pi R_1 \qquad 4\lambda = 2\pi R_4 \; \frac{4\lambda_1}{\lambda_2} = \frac{R_4}{R_1} = \frac{(4)^2}{(1)^2} \Rightarrow \frac{\lambda_4}{\lambda_1} = \frac{4}{1}$

(C)

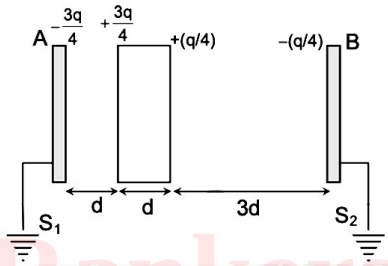


(D) $\frac{T_4}{T_1} = (4)^3 \qquad T \propto n^3 \qquad T = \frac{2\pi r}{v} \Rightarrow n^2 \times n \quad T \propto \frac{n^3}{z^2}$

24. After both the switches S_1 and S_2 are simultaneously closed.

$C_1 = \frac{\varepsilon_0 A}{d} \qquad C_2 = \frac{\varepsilon_0 A}{3d} \qquad C_1 : C_2 = 3:1$

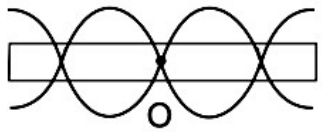
Hence, $q_1 = \frac{3q}{4}$ and $q_2 = \frac{q}{4}$



The charge flown through the switch $S_1, \Delta Q_1 = 2q - \left(-\frac{3q}{4}\right) = \frac{11q}{4}$ (from the plate A to the earth)

The charge flown through the switch $S_2, \Delta Q_2 = -\frac{q}{4} - (-q) = \frac{3q}{4}$ (from the earth to the plate B)

25. Velocity of longitudinal wave in the solid rod,



$$v_r = \sqrt{\frac{Y_{rod}}{\rho_{rod}}} = \sqrt{\frac{2\times10^{11}}{8\times10^3}} = 5000 \text{ m/s}$$

$$\frac{3\lambda_r}{2} = \ell \Rightarrow \lambda_r = \frac{2\ell}{3}$$

Also, $\frac{\lambda_g}{2} = \Delta\ell \Rightarrow \lambda_g = 2\Delta\ell$

Now, $\frac{v_g}{v_r} = \frac{\lambda_g}{\lambda_r} \Rightarrow \frac{\lambda_g}{\lambda_r} = \frac{2\Delta\ell}{2\ell/3}$

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$$\Rightarrow v_g = \frac{3\Delta\ell}{\ell} v_r$$

$$\sqrt{\frac{\gamma RT}{M}} = \frac{3\times 0.1\times 5\times 10^3}{1.2}$$

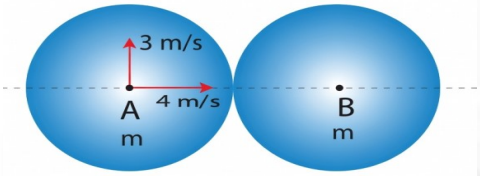
$$\sqrt{\frac{5\times 25\ T}{3\times 3\times 4\times 10^{-3}}} = \frac{5\times 10^3}{4}$$

$$\frac{5\times 25\ T}{9\times 4\times 10^{-3}} = \frac{25\times 10^6}{4\times 4}$$

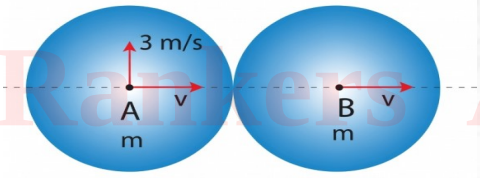
$$T = \frac{9000}{20} = 450\ K$$

26. **Conceptual**

27. At the maximum deformation, the components of velocities of the balls along the line of impact must be equal.



Just before collision



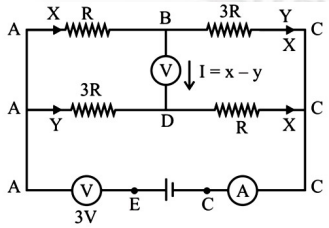
At the maximum deformation

Using the conservation of momentum of the system,

$$mv_0\cos 37 = 2mv \qquad \Rightarrow v = \frac{2v_0}{5} = 2\ m/s$$

The Kinetic energy of the striking ball that has tumed into elastic potential energy at the maximum deformation $= \frac{1}{2}\times 1\left[5^2 - \left(3^2 + 2^2 + 2^2\right)\right] = 4\ J$

28.



$x + y = 6 \Rightarrow x = 3.5\ mA \quad x = 1 + y\ y = 2.5\ mA$

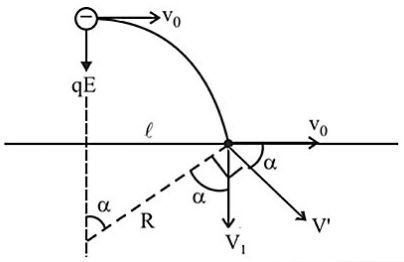
KVL : $\quad -3.5R - 500 + 2.5\times 3R = 0$

$\Rightarrow 4R = 500 R = 125\Omega$

29. $x = \frac{I_{C.M.}}{mR} = \frac{R}{2}$

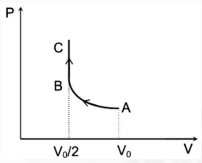
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30.



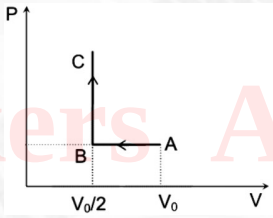
$$\sin\alpha = \frac{v_1}{v'} = \frac{\ell}{R} \qquad \frac{\frac{qE}{m}\cdot t}{v'} = \frac{v_0 t}{\frac{mv'}{qB}} = \frac{v_0 tqB}{m} \qquad E = v_0 B \Rightarrow B = \frac{E}{v_0}$$

31. (P) $P_0V_0 = \frac{P_BV_0}{2} \Rightarrow P_B = 2P_0$



$$B \to C, Q = \frac{3}{2}nR(T_C - T_B)$$

$$P_0V_0 = \frac{3}{2}\left(P_C\frac{V_0}{2} - 2P_0\frac{V_0}{2}\right) \qquad P_C = \frac{10}{3}P_0$$



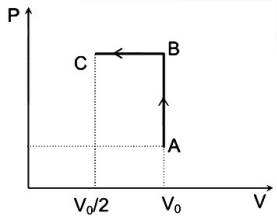
(Q) $\frac{P_0V_0}{T_0} = \frac{P_0\frac{V_0}{2}}{T_B}$

$$\Rightarrow T_B = \frac{T_0}{2} \qquad Q = \frac{3}{2}nR(T_C - T_B)$$

$$P_0V_0 = \frac{3}{2}\left(P_C\frac{V_0}{2} - P_0\frac{V_0}{2}\right)$$

$$P_0 = \frac{3}{2}\left(\frac{P_C}{2} - \frac{P_0}{2}\right) \qquad P_C = \frac{7P_0}{3}$$

(R) $Q = \frac{3}{2}nR(T_B - T_0)$

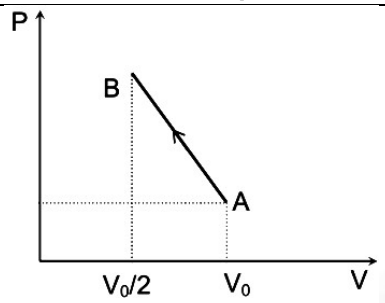


$$P_0V_0 = \frac{3}{2}(P_BV_0 - P_0V_0) \qquad P_B = \frac{5}{3}P_0$$

(S) $Q + W_{ex} = \frac{3}{2}nR(T_B - T_A)$

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$$P_0V_0+\frac{1}{2}\frac{V_0}{2}(P_B+P_0)=\frac{3}{2}\left(P_B\frac{V_0}{2}-P_0V_0\right) \qquad P_B=\frac{11}{2}P_0$$

32. Where displacement amplitude is maximum → Pressure variation is minimum (Pressure node)

- Where displacement amplitude is minimum → Pressure variation is maximum (Pressure antinode)
- P: Magnitude of change in pressure is maximum
  → Pressure amplitude is maximum → $|\cos(4\pi x)|=1\rightarrow 1,2,3$
- Q: Magnitude of change in pressure is minimum
  → Pressure amplitude is minimum → $|\cos(4\pi x)|=0\rightarrow 4,5$
- R: Maximum amplitude of vibration
  → Displacement amplitude is maximum → $|\sin(4\pi x)|=1\rightarrow 4,5$
- S: Minimum amplitude of vibration
  → Displacement amplitude is minimum → $|\sin(4\pi x)|=0\rightarrow 1,2,3$

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33. $T=2\pi\sqrt{\frac{\ell}{g}}\Rightarrow g=\frac{4\pi^2\ell}{T^2}$

$$\frac{\Delta g}{g}=\frac{\Delta\ell}{\ell}+2\frac{\Delta T}{T} \qquad \left[\frac{\Delta T}{T}=\frac{\Delta t/n}{t/n}=\frac{\Delta t}{t}\right]$$

$$\frac{\Delta g}{g}=\frac{\Delta\ell}{\ell}+2\left(\frac{\Delta t}{t}\right)$$

34. The charge +q is given to the inner conducting sphere. Since it is a conductor, the entire charge resides on its outer surface. **Charge on A = +q**

The dielectric shell (inner radius 2a, outer radius 3a) gets polarized due to the electric field of charge +q.

For a dielectric shell with dielectric constant **k** and free charge +q inside it:

Bound charge on inner surface **B** = **–q (1 – 1/k)**

Bound charge on outer surface **C** = **+q (1 – 1/k)**

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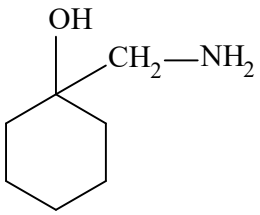
35. P (2.18 on Pauling's scale, 2.1 mentioned in NCERT)
Si (1.8), Sn (1.8), Sr (s-block element, highly electropositive, ≈ 1), Se (**2.55**)
Refer NCERT Class XI, Table 11.3 and NCERT Class XII, Table 7.6 shown below for data on electronegativity values.

Property		Element				
		Carbon C	Silicon Si	Germanium Ge	Tin Sn	Lead Pb
Atomic Number		6	14	32	50	82
Atomic mass (g mol^{-1})		12.01	28.09	72.60	118.71	207.2
Electronic configuration		$[He]2s^2 2p^2$	$[Ne]3s^2 3p^2$	$[Ar]3d^{10}4s^2 4p^2$	$[Kr]4d^{10}5s^2 5p^2$	$[Xe]4f^{14}5d^{10}6s^2 6p^2$
Covalent radius/pm[a]		77	118	122	140	146
Ionic radius M^{4+}/pm[b]		–	40	53	69	78
Ionic radius M^{2+}/pm[b]		–	–	73	118	119
Ionization enthalpy/ kJ mol^{-1}	$\Delta_i H_1$	1086	786	761	708	715
	$\Delta_i H_2$	2352	1577	1537	1411	1450
	$\Delta_i H_3$	4620	3228	3300	2942	3081
	$\Delta_i H_4$	6220	4354	4409	3929	4082
Electronegativity[c]		2.5	1.8	1.8	1.8	1.9
Density[d]/g cm^{-3}		3.51[e]	2.34	5.32	7.26[f]	11.34
Melting point/K		4373	1693	1218	505	600
Boiling point/K		–	3550	3123	2896	2024
Electrical resistivity/ ohm cm (293 K)		10^{14}–10^{16}	50	50	10^{-5}	2×10^{-5}

Property	O	S	Se	Te	Po
Atomic number	8	16	34	52	84
Atomic mass/g mol^{-1}	16.00	32.06	78.96	127.60	210.00
Electronic configuration	$[He]2s^2 2p^4$	$[Ne]3s^2 3p^4$	$[Ar]3d^{10}4s^2 4p^4$	$[Kr]4d^{10}5s^2 5p^4$	$[Xe]4f^{14}5d^{10}6s^2 6p^4$
Covalent radius/(pm)[a]	66	104	117	137	146
Ionic radius, E^{2-}/pm	140	184	198	221	230[b]
Electron gain enthalpy, $/\Delta_{eg}H$ kJ mol^{-1}	–141	–200	–195	–190	–174
Ionisation enthalpy ($\Delta_i H_1$) /kJ mol^{-1}	1314	1000	941	869	813
Electronegativity	3.50	2.58	2.55	2.01	1.76
Density /g cm^{-3} (298 K)	1.32[c]	2.06[d]	4.19[e]	6.25	–
Melting point/K	55	393[f]	490	725	520
Boiling point/K	90	718	958	1260	1235
Oxidation states*	–2,–1,1,2	–2,2,4,6	–2,2,4,6	–2,2,4,6	2,4

*Single bond; [b]Approximate value; [c]At the melting point; [d] Rhombic sulphur; [e]Hexagonal grey; [f]Monoclinic form, 673 K.
* Oxygen shows oxidation states of +2 and +1 in oxygen fluorides OF_2 and O_2F_2 respectively.

36. X =



Structure of X contains **tertiary alcohol** and primary amine as the functional groups.

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37. CO_2 (sublimation point $-78^{\circ}C$ at 1 atm), NO_2 (boiling point $21^{\circ}C$) and SO_2 (boiling point $-10^{\circ}C$) are **gases at room temperature**

38. When $\alpha = 0$ (no dissociation)

Vapour density $= \frac{MW_{SO_3}}{2} = \frac{80}{2} = \mathbf{40}$

When $\alpha = 1$ (complete dissociation)

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

initial moles	1	0	0
final moles	0	1	0.5

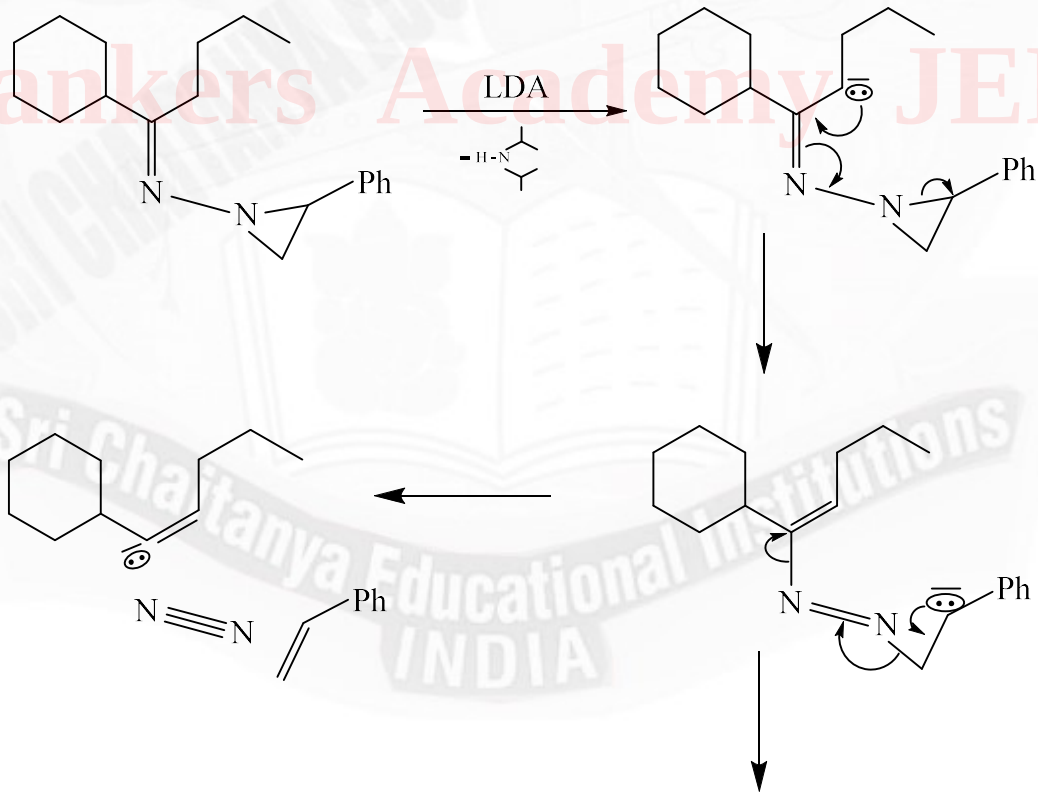
$$\text{Average molecular weight} = \frac{MW_{SO_2} + MW_{O_2} \times 0.5}{1 + 0.5}$$

$$= \frac{64 \times 1 + 32 \times 0.5}{1.5} = \frac{80}{1.5} = \frac{80 \times 2}{3} = 53.33$$

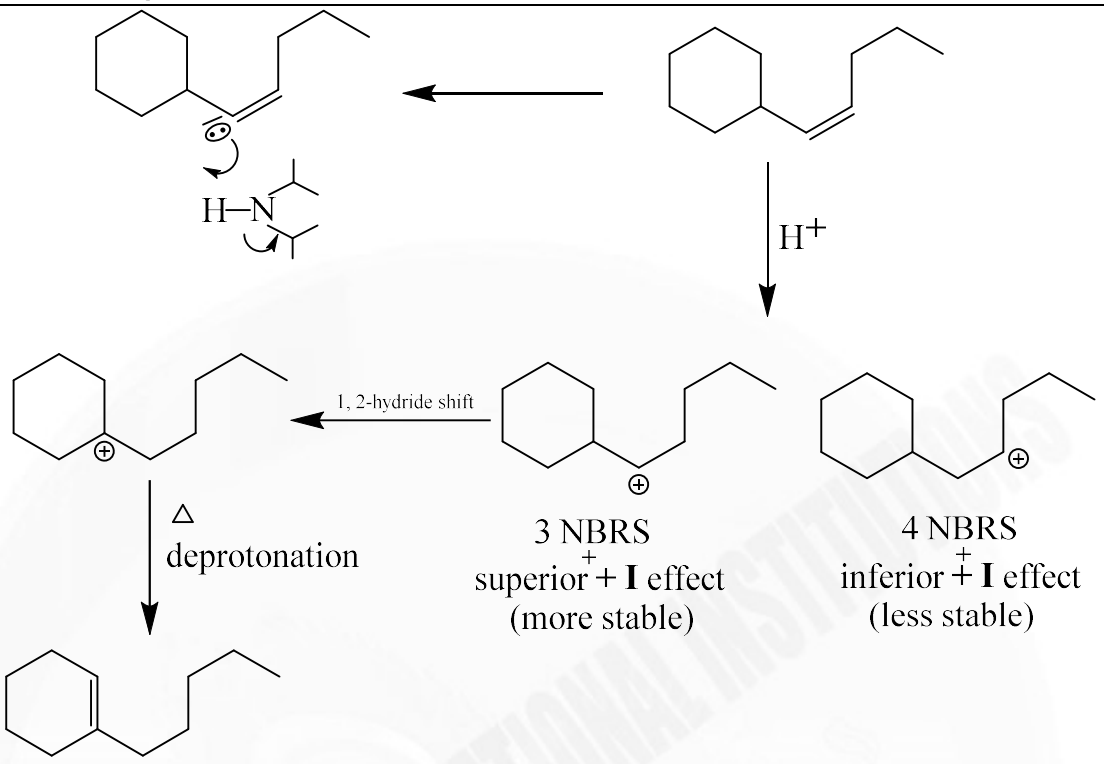
Vapour density $= \frac{M}{2} = \mathbf{26.66}$

Vapour density of the equilibrium mixture $(0 < \alpha < 1)$ must lie between these extremes (between 26.66 and 40)

39.



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40. I (sucrose) is a high calorie sweetener (4 kcal/g) because it is metabolized by the human body.
II (sucralose) is a zero calorie sweetener (0 kcal/g) because it is excreted without being broken down.
Refer Class XII – Surface chemistry (enzymes), Biomolecules (carbohydrates) and Chemistry in everyday life (artificial sweetening agents) for other details.

41. AgF and $CaCl_2$ are soluble in water

$AgCl\ (K_{sp} = 1.8 \times 10^{-10})$ and $CaF_2\ (K_{sp} = 5.3 \times 10^{-9})$ are precipitates or sparingly soluble salts.

$AgCl + 2NH_3 \longrightarrow [Ag(NH_3)_2]^+ Cl^-$ (linear complex **cation**)

Refer lab manual, Unit-7, Page – 88 for Hazard Warning (shown below)

Hazard Warning

- *The solution obtained after dissolving AgCl/AgBr/AgI precipitate in ammonium hydroxide should be acidified with 2M HNO_3 and should be discarded quickly to avoid serious explosion.*

42. Statements – 1, 3, 4 and 5 are correct.

43. $84 + 84 + 2\theta_1 = 360$

$84 + 94 + 2\theta_2 = 360$

$\theta_1 = 96^\circ \qquad \theta_2 = 91^\circ \qquad m = 8$

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$(\theta_1 - \theta_2) \times m = (96-91) \times 8 = 5 \times 8 = \mathbf{40}$

44. $E_{cell} = E^{\circ}_{cell} - \dfrac{2.303\,RT}{nF}\log Q_c$

$H^+ \left| \dfrac{1}{2}H_2 \right\| Ag^+ \left| Ag \right.$

$0.709 = 0.799 - \dfrac{0.06}{1}\log\dfrac{1}{[Ag^+]} \qquad -0.09 = 0.06\log[Ag^+]$

$[Ag^+] = 10^{-3/2}\,M$

$Ag_2SO_4 \rightleftharpoons 2Ag^+ + SO_4^{2-}$

S moles/lit $\quad$ 2S $\quad$ S

$K_{sp} = 4s^3 \quad 2S = 10^{-3/2}$

$4S^2 = 10^{-3};\ S = \dfrac{10^{-3/2}}{2}$

$4S^3 = \dfrac{10^{-9/2}}{2} = \dfrac{10^{-1/2}}{2}\times 10^{-4} = \dfrac{1}{2\sqrt{10}}\times 10^{-4}$

$= \dfrac{1}{2}\cdot\dfrac{\sqrt{10}}{10}\times 10^{-4} \qquad = \dfrac{1}{2}\cdot\dfrac{3.16}{10}\times 10^{-4} \qquad = \dfrac{0.316}{2}\times 10^{-4}$

$= 0.158\times 10^{-4} \qquad = \boxed{1.58\times 10^{-5}}$

$x = 1$
$y = 5$
$z = 8$

$\sqrt{x-y+z} = \sqrt{1-5+8} \qquad = \sqrt{4} = \mathbf{2}$

45. $CaO + 3C \longrightarrow CaC_2 + CO$

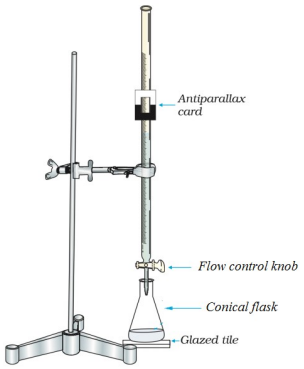
1000 kg crude product contains 640 kg CaC_2 and 360 kg CaO (which did not react)

$n_{CaC_2} = \dfrac{640\times 10^3}{64} = 10^4$ moles $\ \therefore\ n_{CaO} = 10^4$ moles

$w_{CaO} = 56\times 10^4\ g = 560$ kg

Total CaO added = 560 kg (reacted) + 360 kg (unreacted) = **920** kg

46. Refer lab Manual, Unit – 6, figure 6.3 shown below



47.

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Name	M.F	E.F	E.F.W
Hyponitrous acid	$H_2N_2O_2$	HNO	31
$\beta-D$ ribose	$C_5H_{10}O_5$	CH_2O	30
Gammaxene	$C_6H_6Cl_6$	CHCl	47
Metaphosphoric acid	$(HPO_3)_n$	HPO_3	79

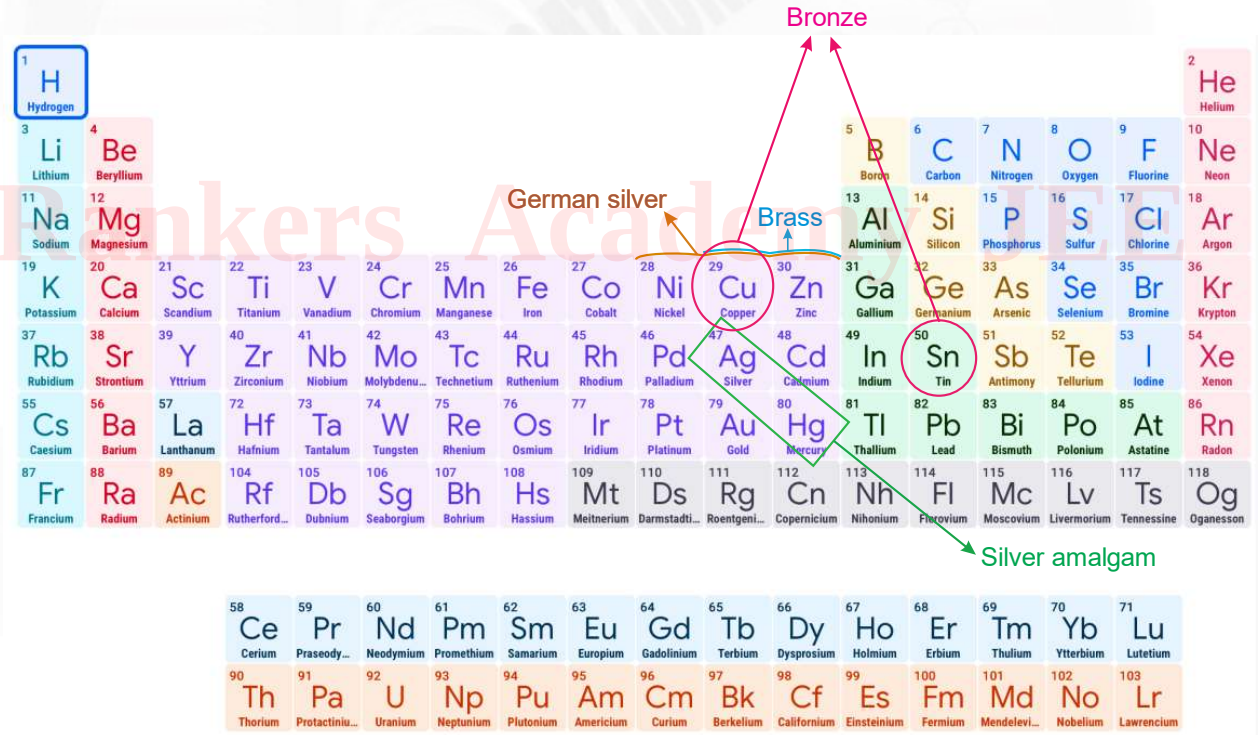
Sum of EFWs $= 31 + 30 + 47 + 79 =$ **187**

48. Perkin reaction produces *cinnamic acid* as the product, not *cinnamaldehyde.*

49. CaO has rock salt structure, chromite $(FeCr_2O_4)$ has spinel structure and Fe_3O_4 has inverse spinel structure.

50. While exhaustive catalytic hydrogenation will destroy the optical activity of the given hydrocarbon and convert it into an alkane (C_nH_{2n+2}); addition of 1 mole H_2 (irrespective of whether it adds across $C=C$ or $C\equiv C$) in presence of Pd catalyst (cis addition) will keep the product optically active with index of hydrogen deficiency dropping down to 2 (from 3 in the reactant given)

51.



Silver amalgam used in dentistry also contains metals like tin, copper & zinc, besides silver and mercury and is hence not a binary solution

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